



MAJOR PROJECT PRESENTATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BARAK VALLEY ENGINEERING COLLEGE
NIRALA, SRIBHUMI
(16-06-2026)

Tablekard: A Smart QR-Based Multi-Tenant Restaurant Management SaaS Platform

TEAM MEMBERS (GROUP NO.3)

Faruk Khan	222010007019
Mridul Roy	222010007027
Sanjeev Iqbal Ahmed	222010007041
Shahid Anowar	222010007043
Sourav Sharma	222010007049

SUPERVISOR
N RANA SINGHA

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INTRODUCTION

Tablekard is a comprehensive, **multi-tenant Software-as-a-Service (SaaS)** platform that enables seamless, contactless QR-based ordering and restaurant management – without expensive Point-of-Sale (POS) hardware.

Feature	Description
No App Download	Customers order via web browser by scanning a QR code
Multi-Tenant SaaS	One platform serves many restaurants with strict data isolation
Real-Time	Orders appear instantly on kitchen/admin dashboards via WebSockets
Secure Payments	Integrated Razorpay gateway (UPI, Cards, Net Banking, Wallets)



PROBLEM STATEMENT

The dine-in sector suffers from high operational friction and outdated manual workflows, necessitating a unified digital infrastructure to address three critical gaps:

1. **Integrated System Gap:** Lack of real-time coordination among customers, kitchen staff, and management leads to slow service, order errors, and limited business insights.
2. **Operational Fragmentation:** Reliance on static menus and disconnected tools causes poor synchronization, communication delays, and kitchen bottlenecks.
3. **Customer Experience & Technology Gap:** Outdated dine-in systems cause delays and ordering uncertainty, creating a need for visual menus, live order tracking, and secure instant payments.



Fig: Wrong order served

OBJECTIVES

To develop a multi-tenant SaaS platform for seamless dine-in management, focusing on:

1. **Multi-Role QR Ordering System:** To develop interconnected, app-free web applications bridging customers, kitchen staff, restaurant owners, and platform admins.
2. **Real-Time Operational Sync:** To enable instant order updates, live queue tracking, and seamless cross-dashboard communication using Supabase real-time features.
3. **Smart & Interactive Dining:** To enhance customer decision-making with AR-based 3D food visualizations and AI-powered personalized recommendations.
4. **Secure SaaS Infrastructure:** To ensure strict multi-tenant data privacy, seamless Razorpay payment integrations, and predictive analytics for business insights.

RELATED WORKS

Author & Year	Research Focus	Key Finding	Relevance to Tablekard
Patil et al. (2025) ^[1]	Browser QR Ordering	Sped up orders 25%; cut errors 90%	Proves browser-based QR efficacy
Gaikar et al. (2026) ^[2]	Real-Time Tracking	Dashboards improve kitchen flow	Validates need for live visual queues
Khare et al. (2023) ^[3]	Diner Perception	77% prefer contactless systems	Proves market demand for no-contact
Xu et al. (n.d.) ^[4]	Tech Acceptance	Web is preferred over downloading apps	Supports our browser-native approach
Styliaras (2021) ^[5]	AR in Hospitality	High engagement, but accessibility is hard	Supports WebXR (no-app) AR strategy

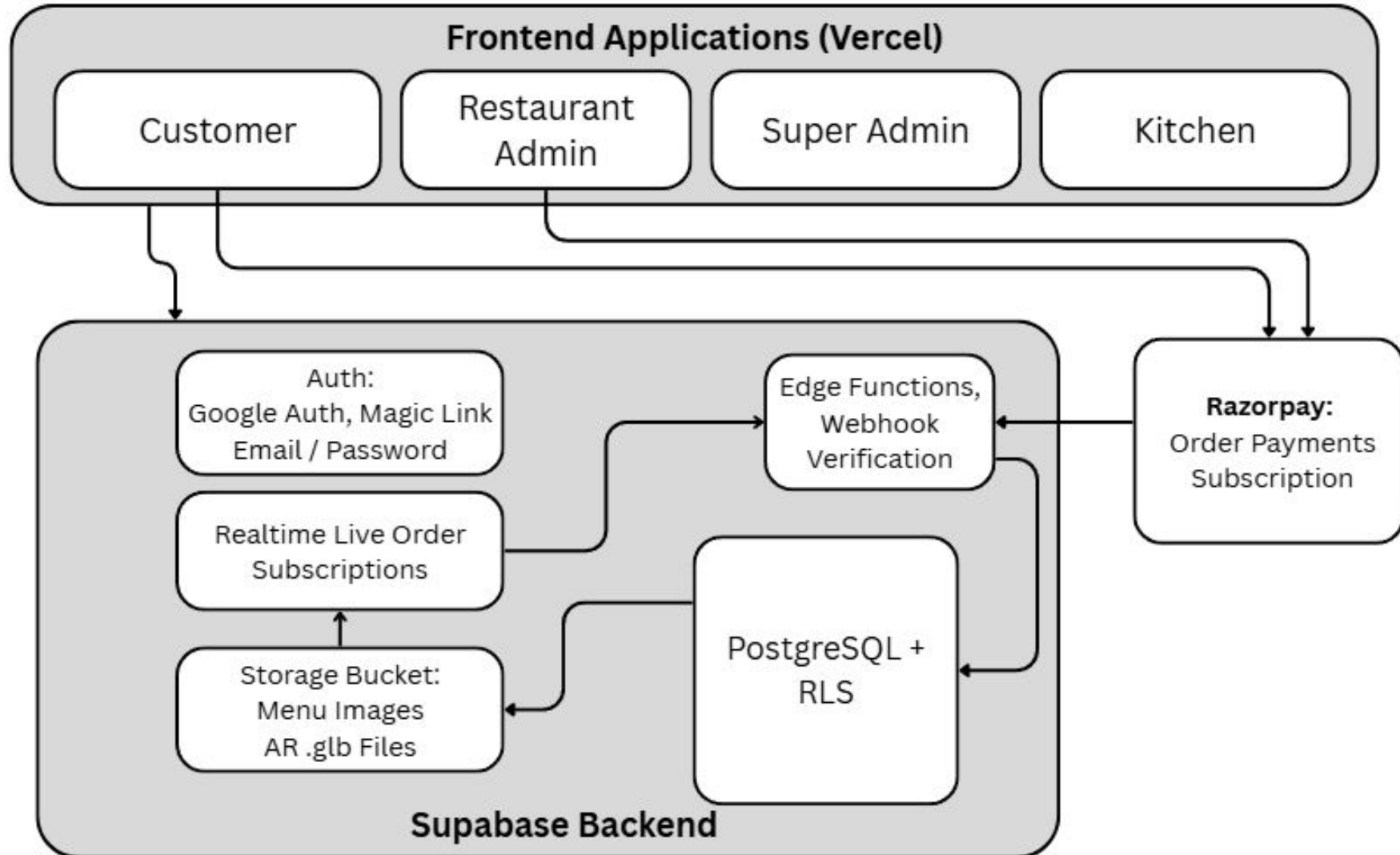
Table: Summary of Academic Research

RELATED WORKS

Platform	Core Tech	Primary Focus	Multi-Tenant	Key Limitation
Toast POS ^[7]	Android, Cloud	Kitchen & Billing	Yes	Expensive proprietary hardware lock-in
OpenTable ^[8]	Web, Native Apps	Reservations	Yes	No direct-to-kitchen QR ordering
Static QR Menus	HTML, PDF	Digital Paper Menus	No	No cart, payment, or live sync
GloriaFood ^[9]	Web Widgets	Takeout / Delivery	No	Lacks AR, AI, and SaaS scalability

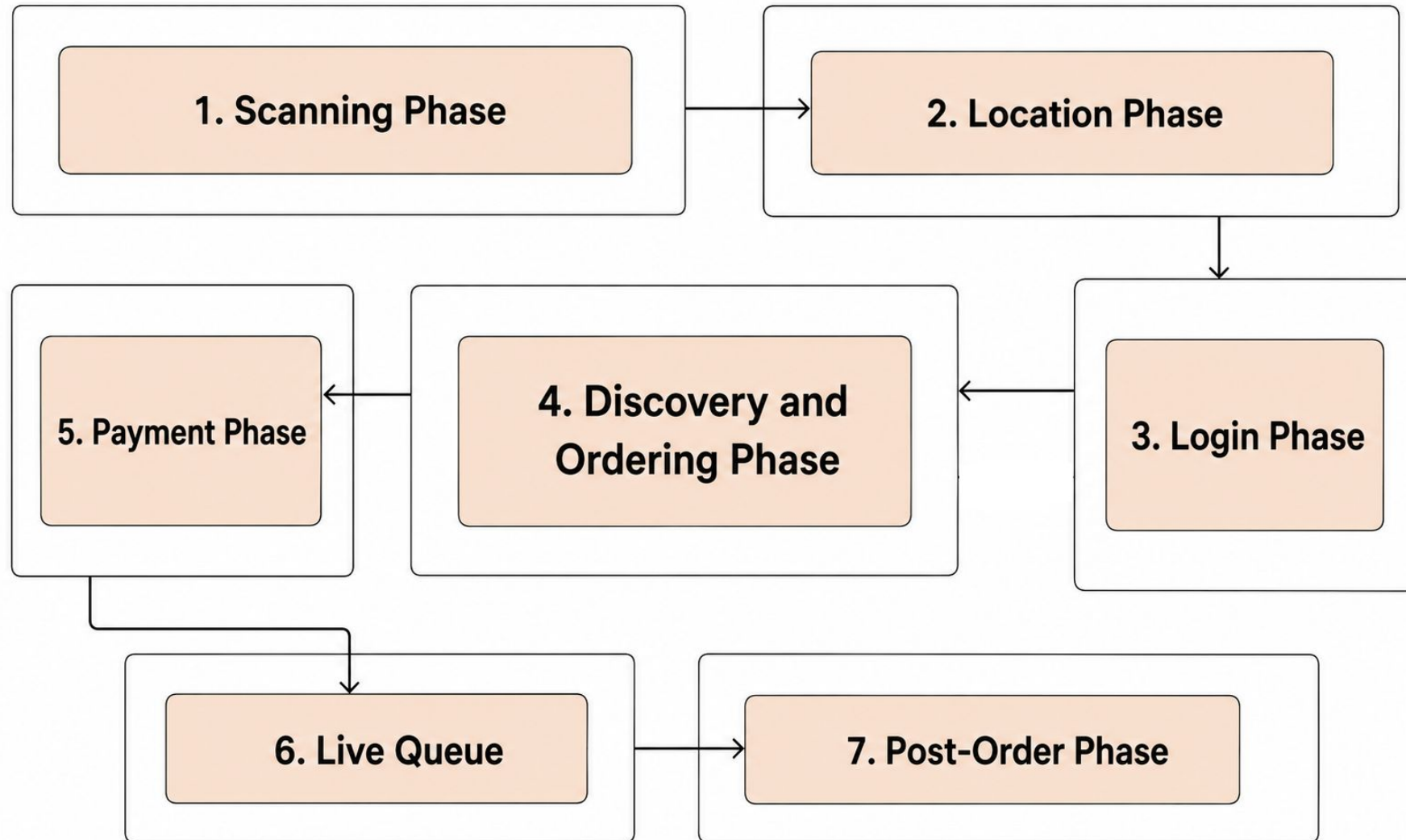
Table: Survey of Existing Commercial Platforms

METHODOLOGY



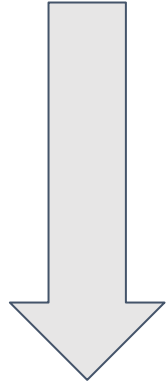
- **Architectural Diagram**

METHODOLOGY

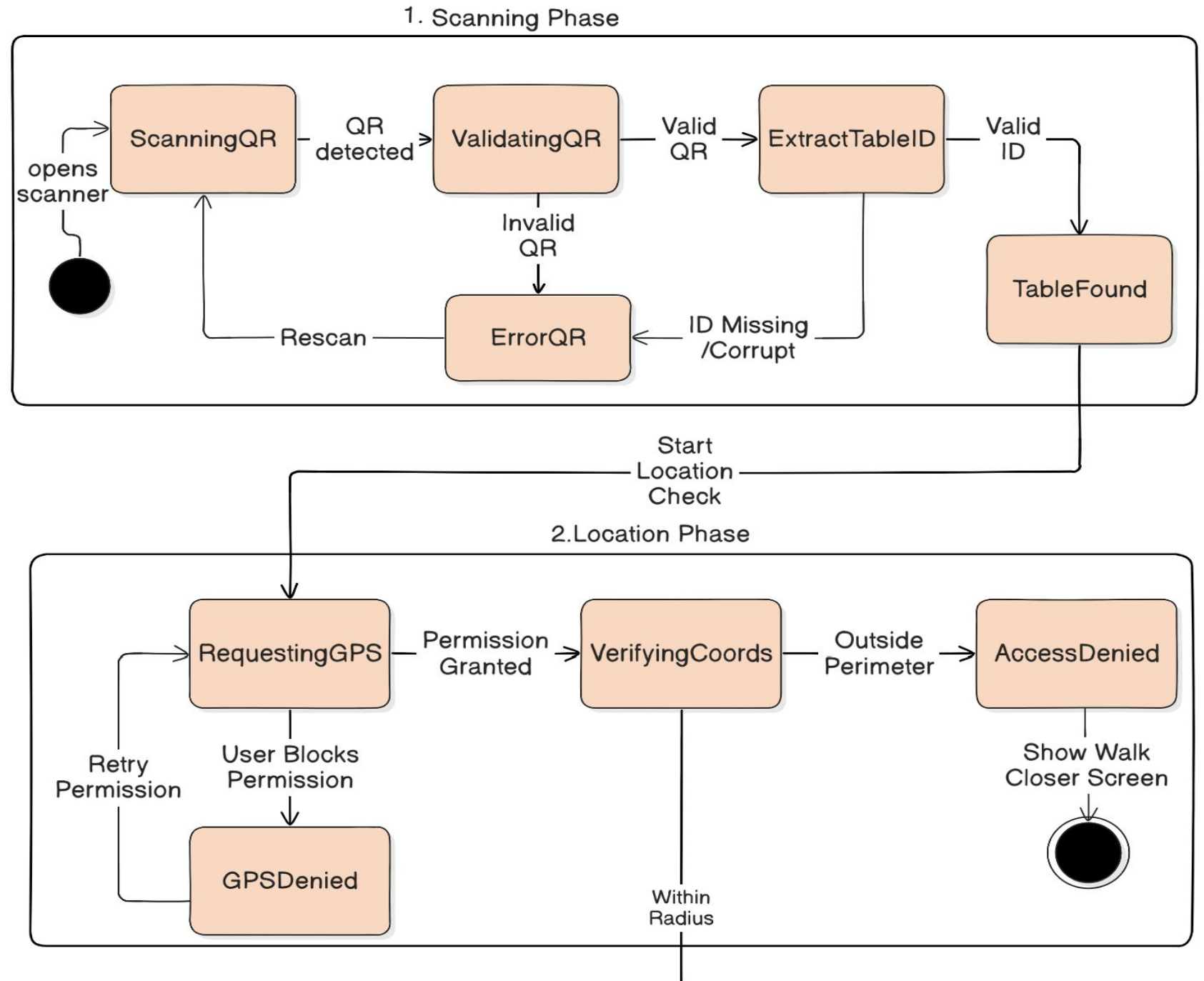


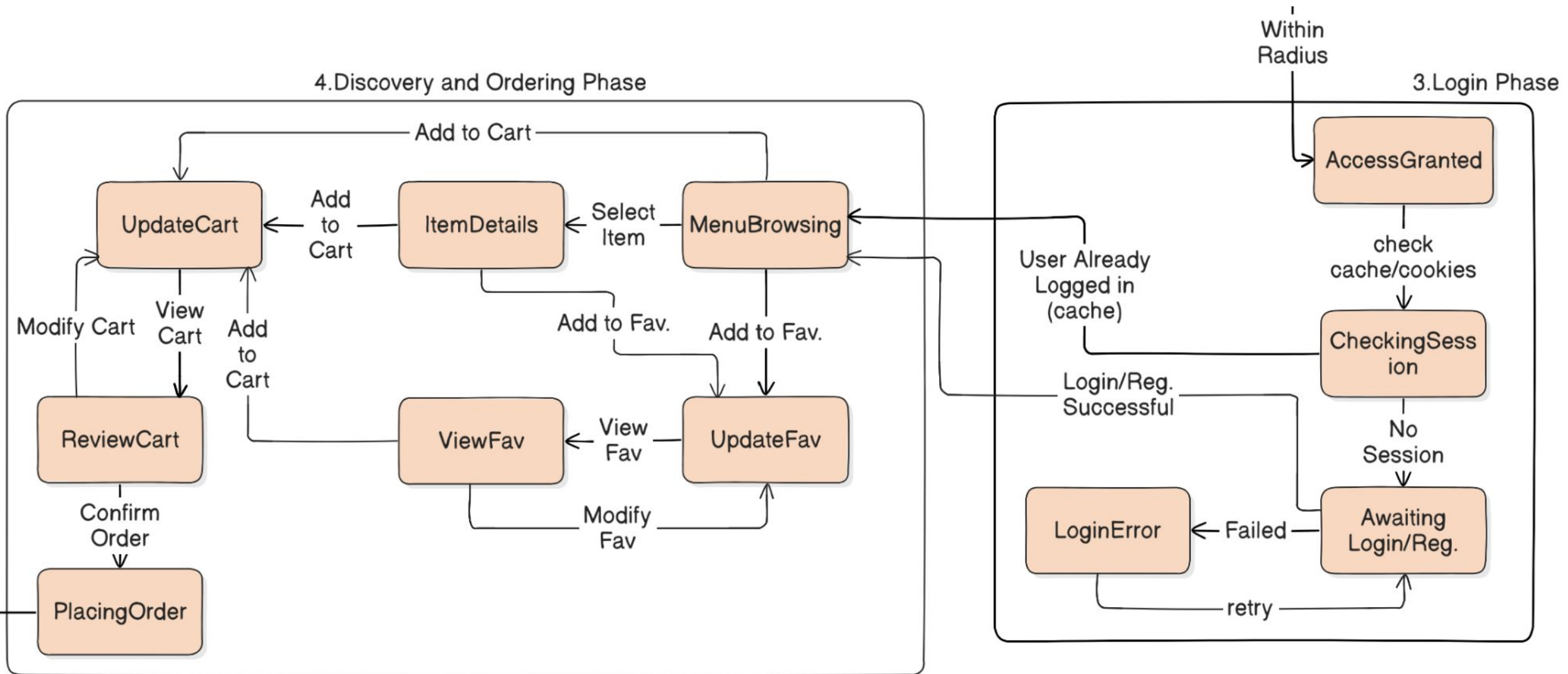
- **State Transition Diagram (overview)**

1. Scanning Phase



2. Location Verification Phase



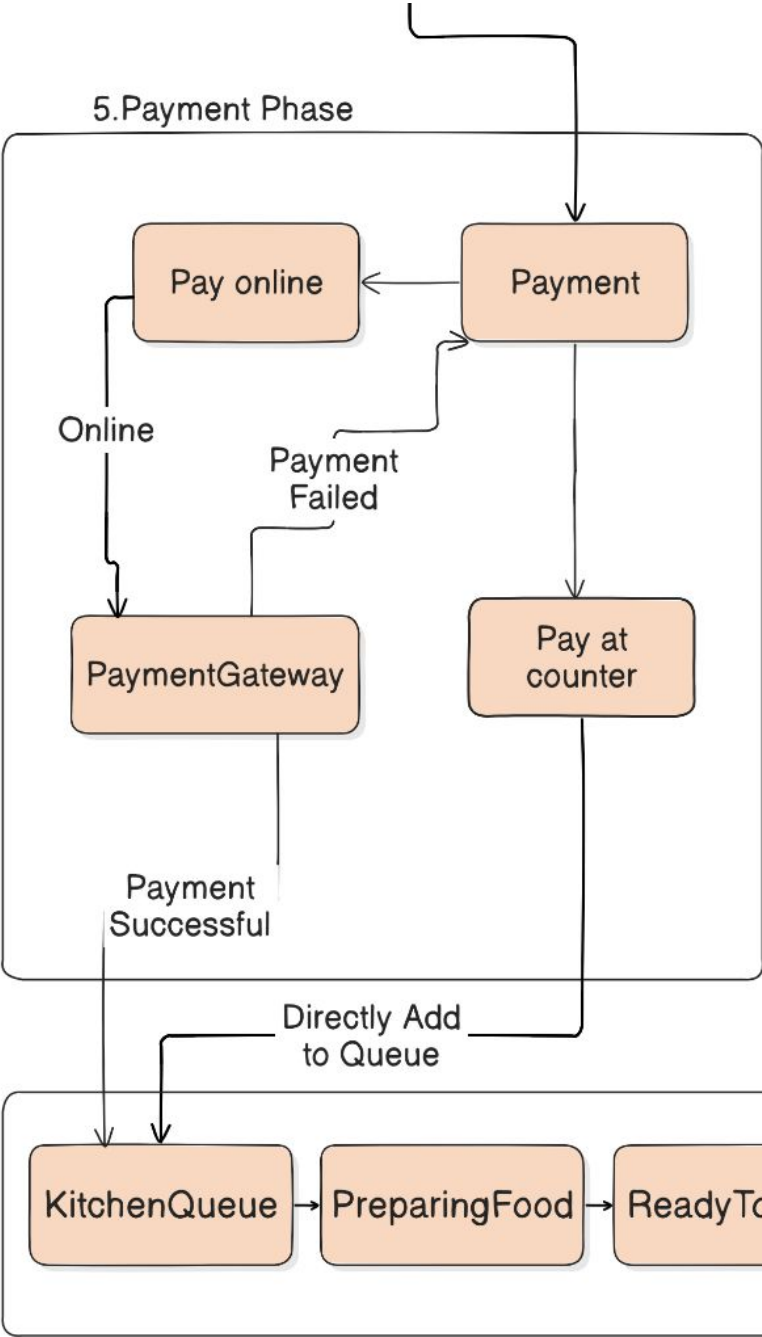
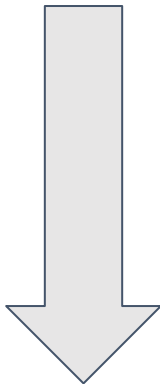


4. Discovery and Ordering Phase

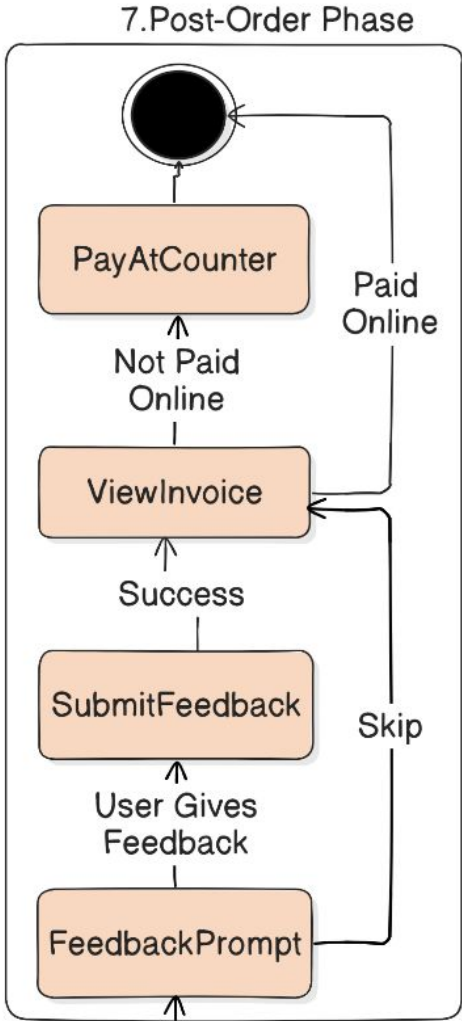


3. Login Phase

5. Payment Phase



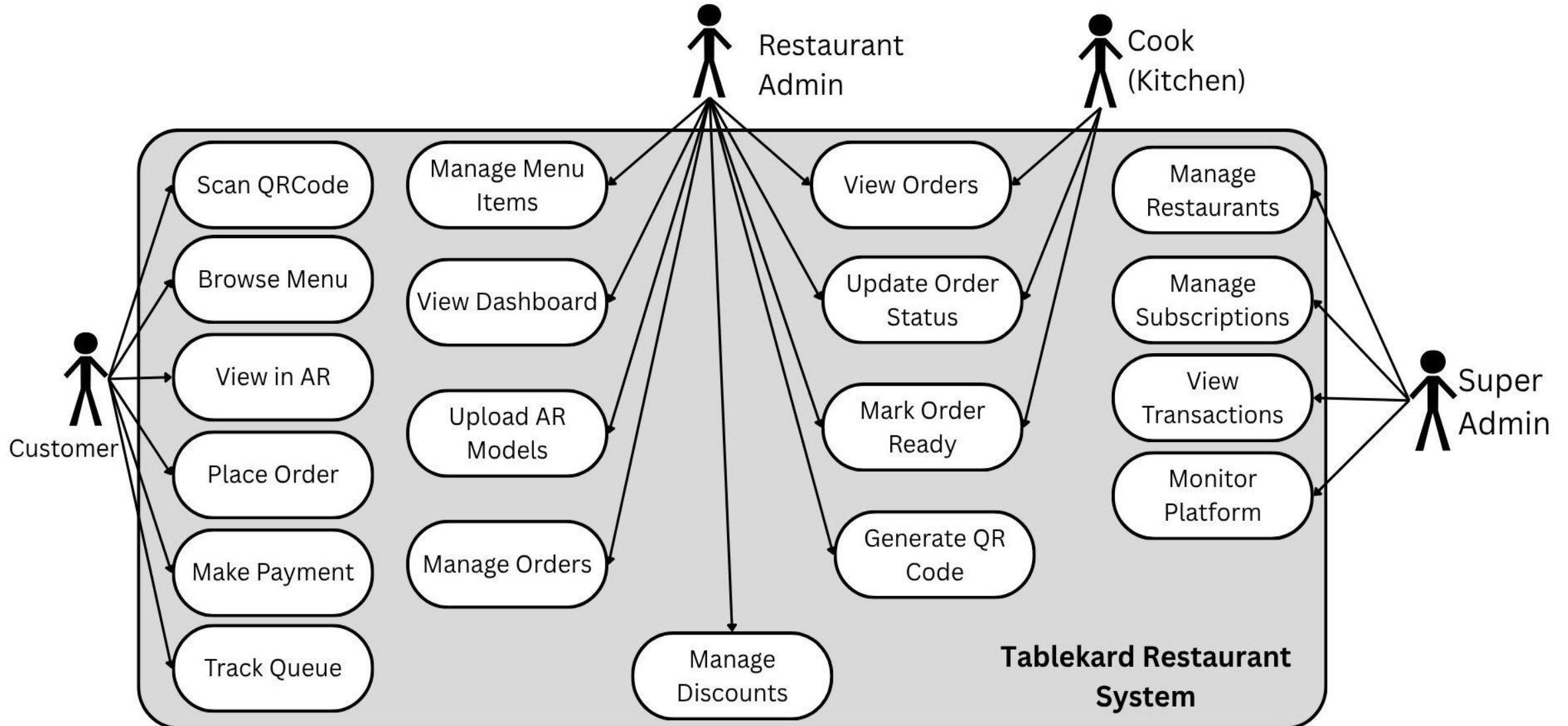
6. Live Queue



7. Post-Order Phase

METHODOLOGY

- Use-Case Diagram



METHODOLOGY

- **Payment Processing (Razorpay Webhooks)**



- SHA-256 signature verification ensures secure transactions.
- Orders are automatically created even if the user's browser session is interrupted.

- **Row-Level Security (RLS)**

- Implemented 35 security policies across 17 database tables.
- Enforces data access control directly at the database layer.

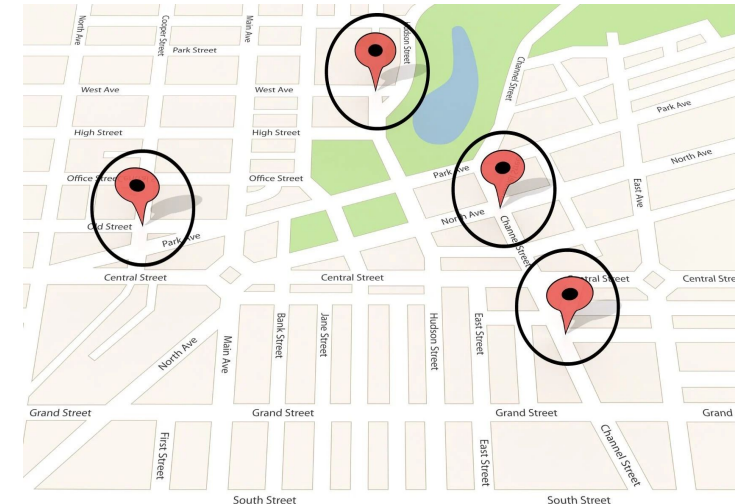
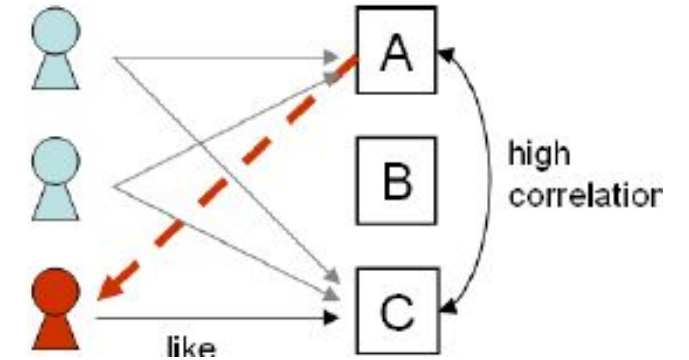


- **Edge Functions**

- Developed 5 server-side functions for secure payment and business logic processing.
- Speeds up response delivery times: networks process requests locally.

METHODOLOGY

- **Item-Based Collaborative Filtering**
 - Recommendation engine built using user preferences, search history, and previous interactions.
 - Generates personalized menu and product suggestions.
- **Geo-Fencing Based Order Validation**
 - Orders can only be placed within a predefined service area.
 - Prevents invalid deliveries and ensures efficient order fulfillment.



REQUIREMENTS

Software Requirements (Development and Production)

Category	Tool/Library	Purpose
Frontend UI	React 19 & Vite	Multi-role SPA (Customer, Admin, Super Admin)
Language	TypeScript	Compile-time type safety
QR Code	qrcode.react	Table QR code generation
Database & Storage	Supabase (PostgreSQL)	Relational data with strict RLS (tenant isolation)
Backend Runtime	Deno (Supabase Functions)	Serverless TypeScript webhook & payment handlers
Authentication	Supabase Auth	Google OAuth, Magic Link, email/password
Payment Gateway	Razorpay SDK & API	Secure checkout integration (UPI, cards, netbanking) and Deno HMAC verification
Realtime	Supabase Realtime	WebSocket-based live order notifications

REQUIREMENTS

Hardware Requirements

Component	Minimum	Recommended
Development Machine	Intel i5 8th Gen / Ryzen 5, 8 GB RAM, 20 GB SSD, Windows 10 / macOS 12 / Ubuntu 20.04	16 GB RAM for ML pipelines; NVIDIA CUDA GPU
Customer Device	Any smartphone/tablet with Chrome 79+, Safari 14+, or equivalent; active mobile data or Wi-Fi	—
Admin Devices	Desktop/laptop, Chrome 90+ / Edge / Firefox / Safari, 1280×720 display, stable broadband	1920×1080 for analytics dashboard

IMPLEMENTATION AND RESULTS

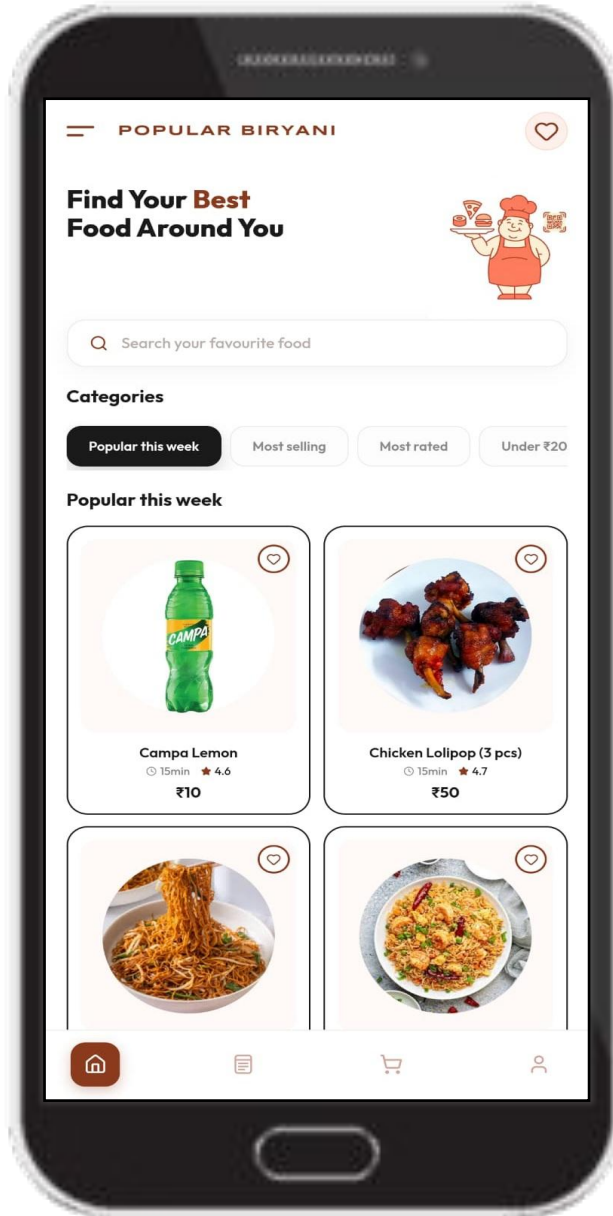


Fig: Tablekard User Web App (Home Screen)

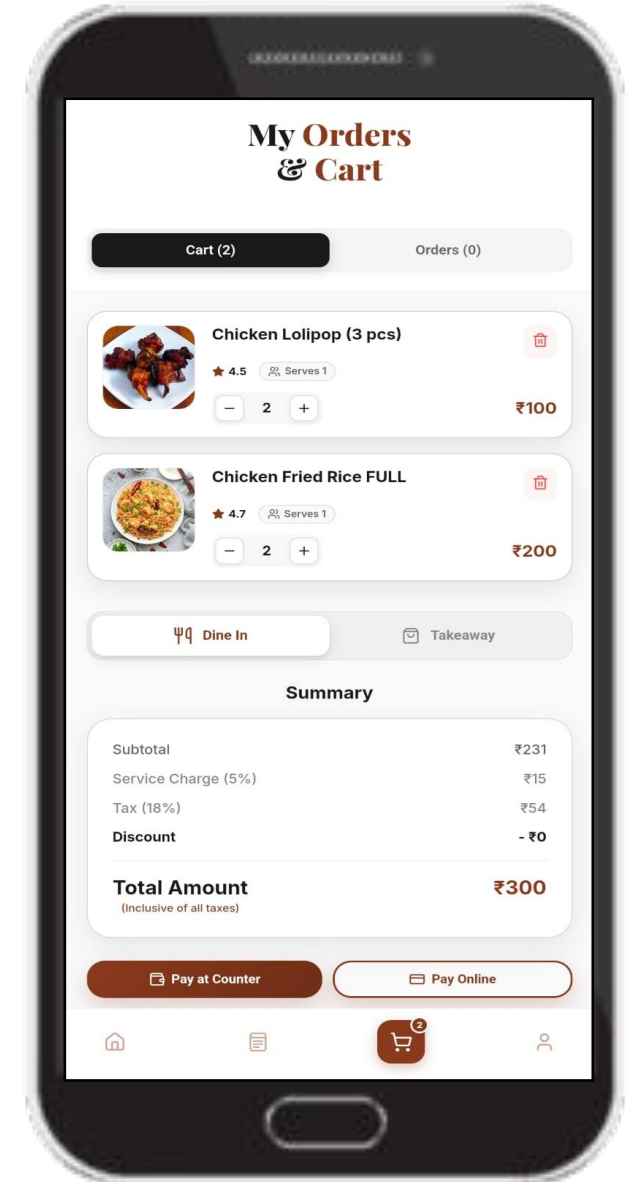


Fig: User Web App (Cart)

IMPLEMENTATION AND RESULTS

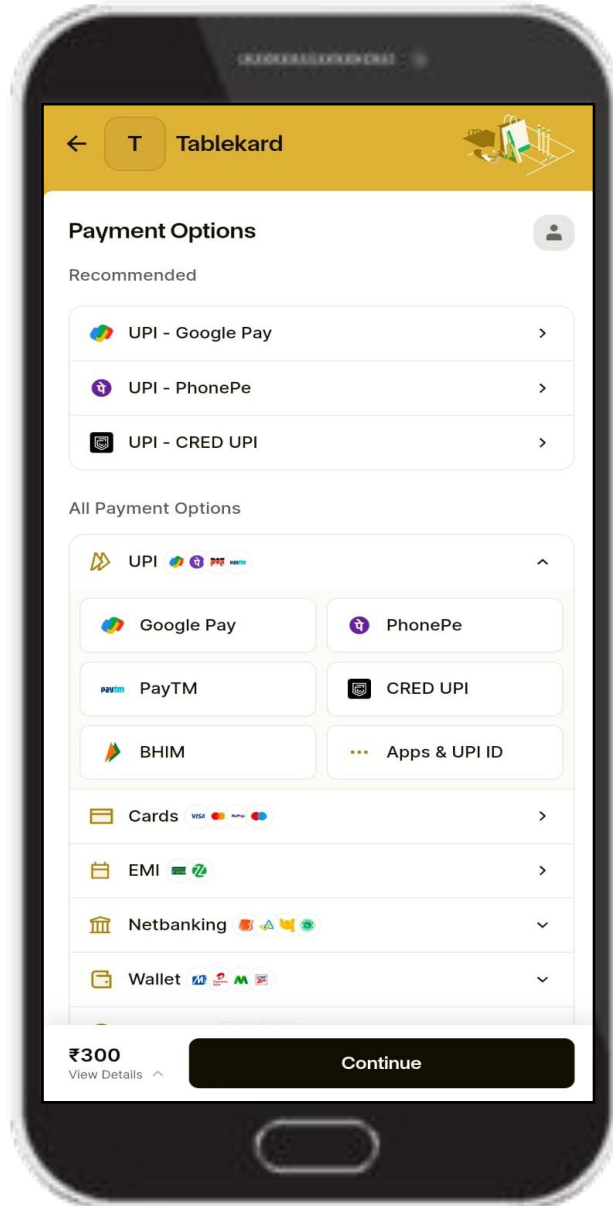


Fig: Razorpay Payment Screen

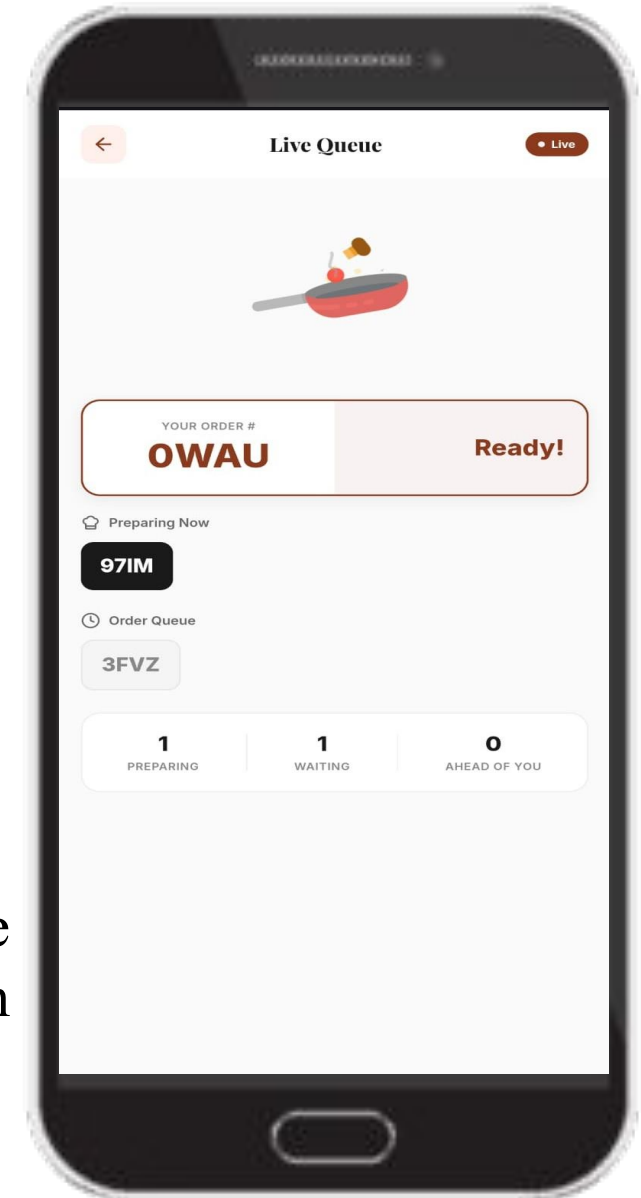


Fig: Order Queue
Screen

IMPLEMENTATION AND RESULTS

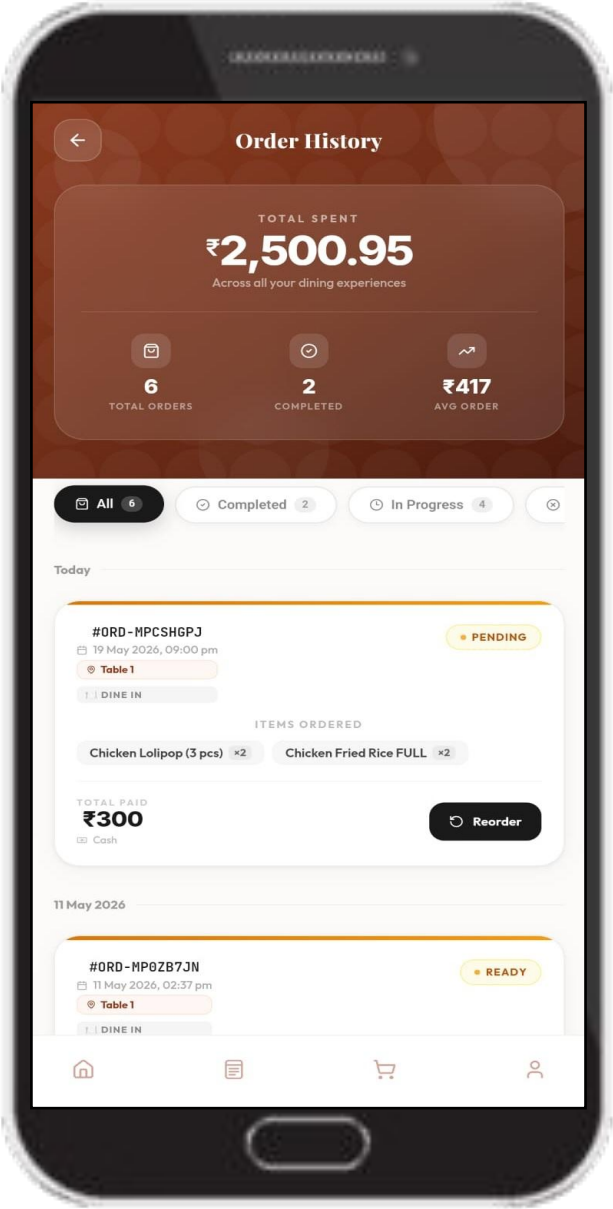


Fig: User Order History

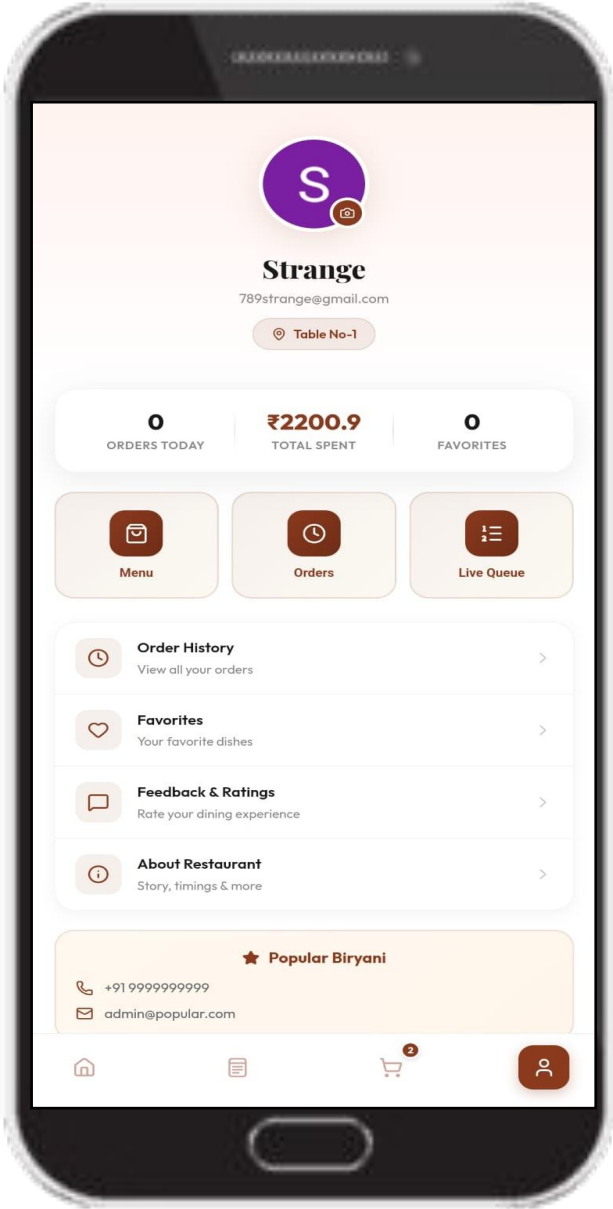


Fig: User Profile

IMPLEMENTATION AND RESULTS

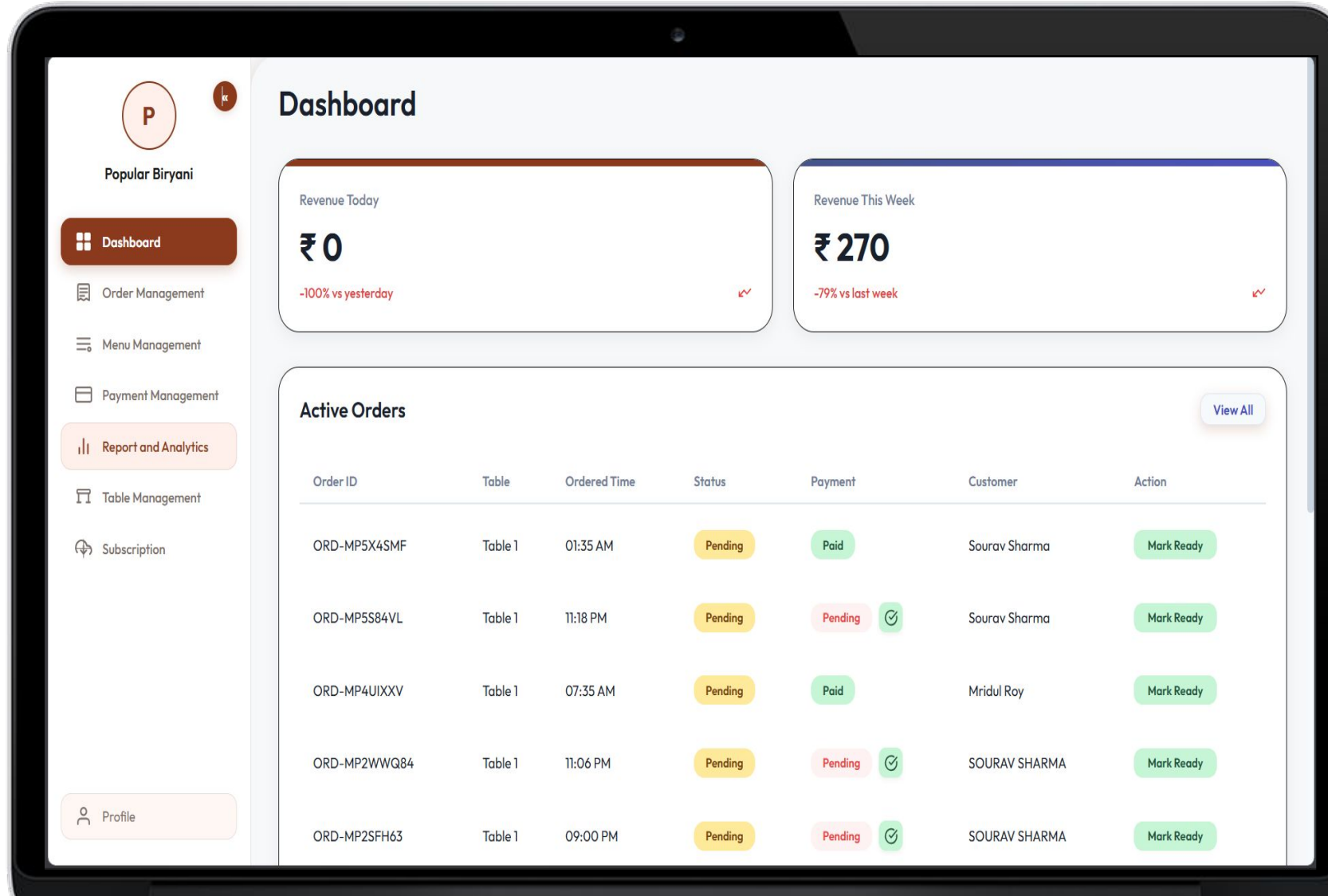


Fig: Restaurant Admin Dashboard

IMPLEMENTATION AND RESULTS

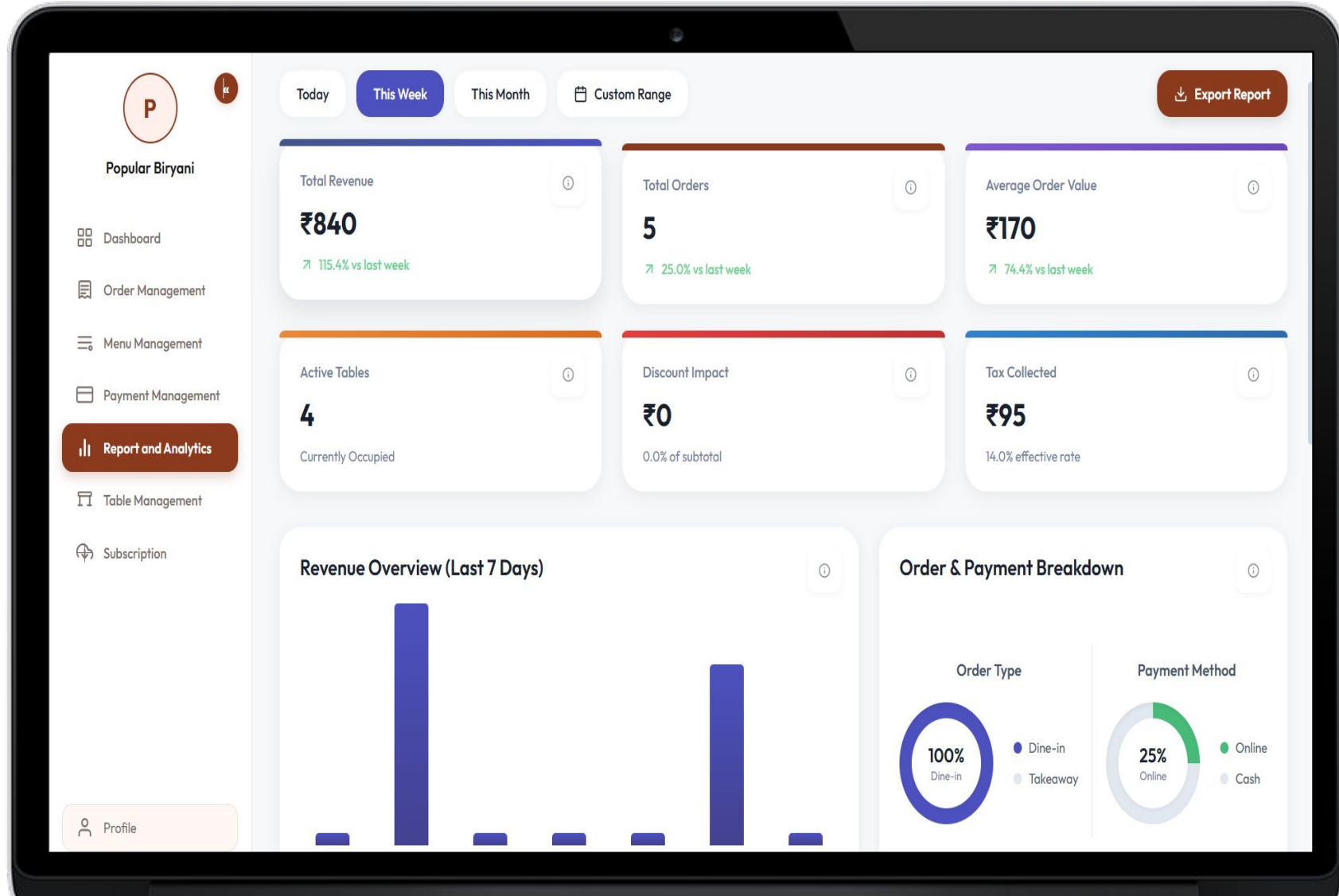


Fig: Restaurant Admin Analytics

IMPLEMENTATION AND RESULTS

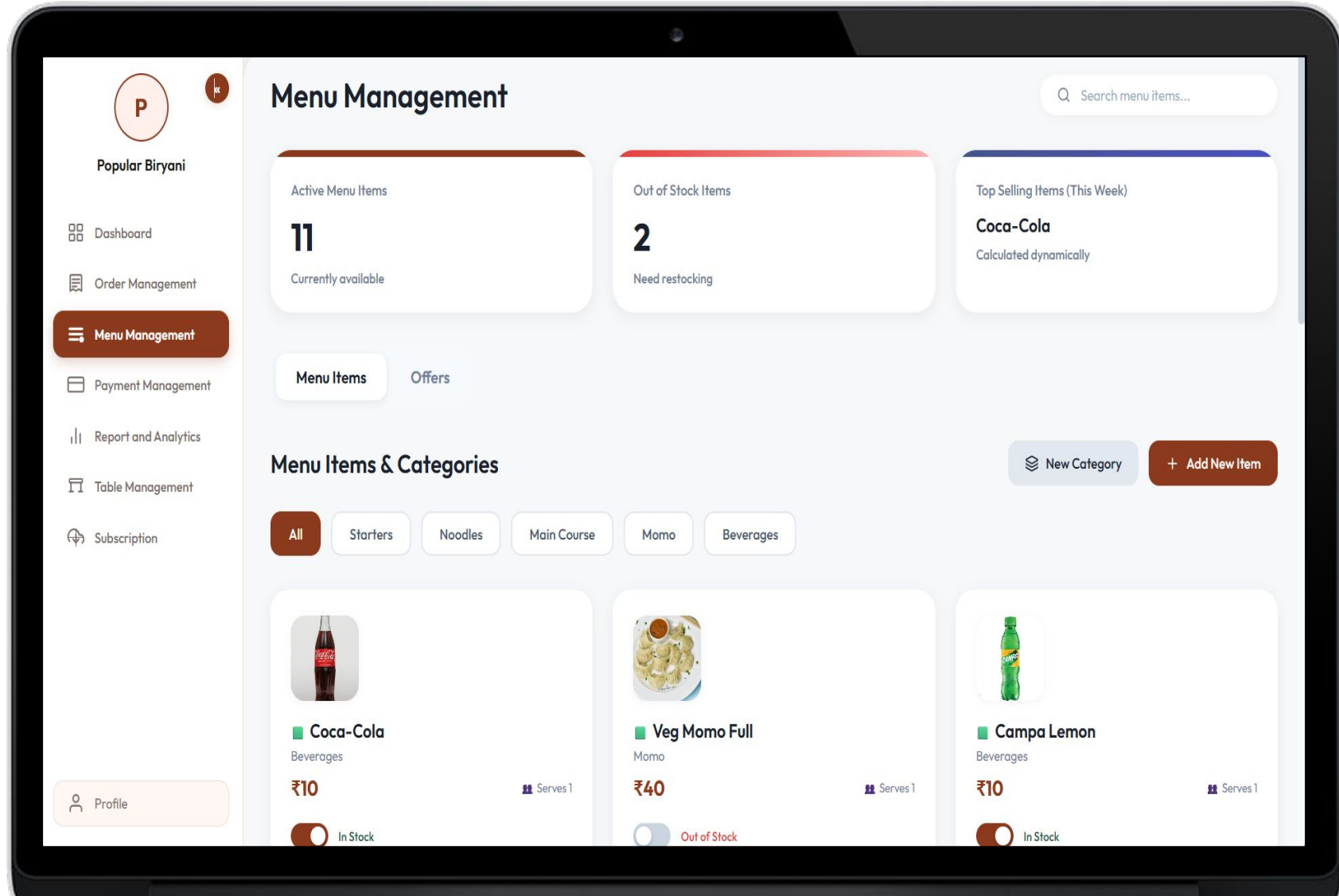


Fig: Restaurant Admin - Menu Management

IMPLEMENTATION AND RESULTS

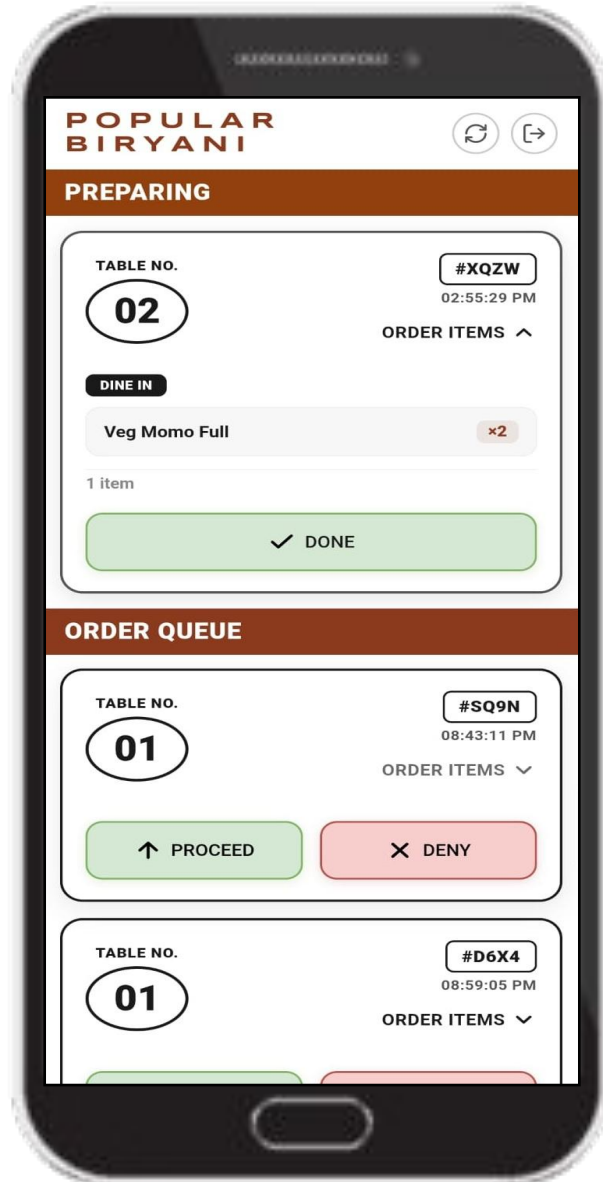


Fig: Restaurant Cook App

IMPLEMENTATION AND RESULTS

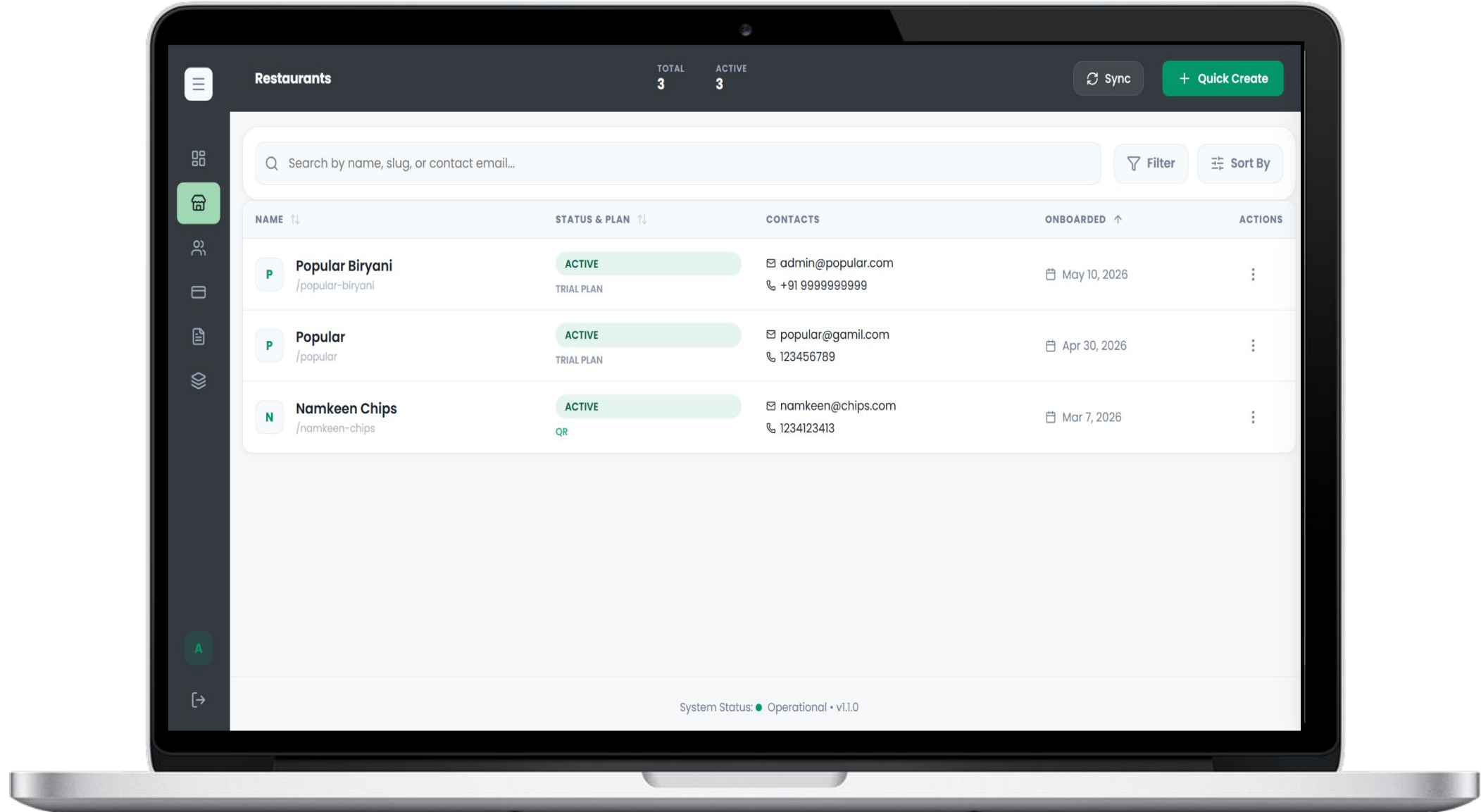


Fig: Super-admin Restaurant management

IMPLEMENTATION AND RESULTS

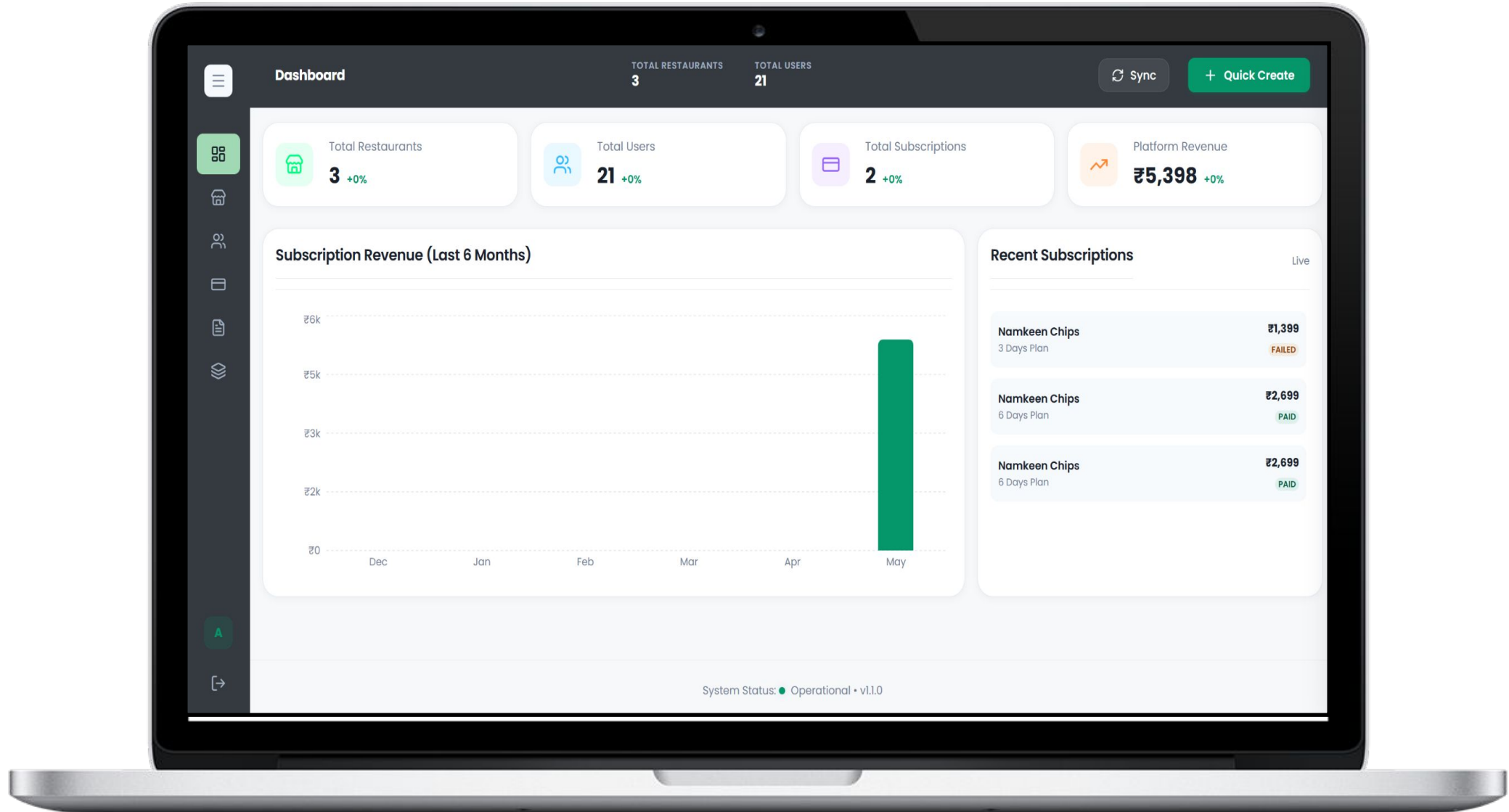


Fig: Super-admin Dashboard

IMPLEMENTATION AND RESULTS

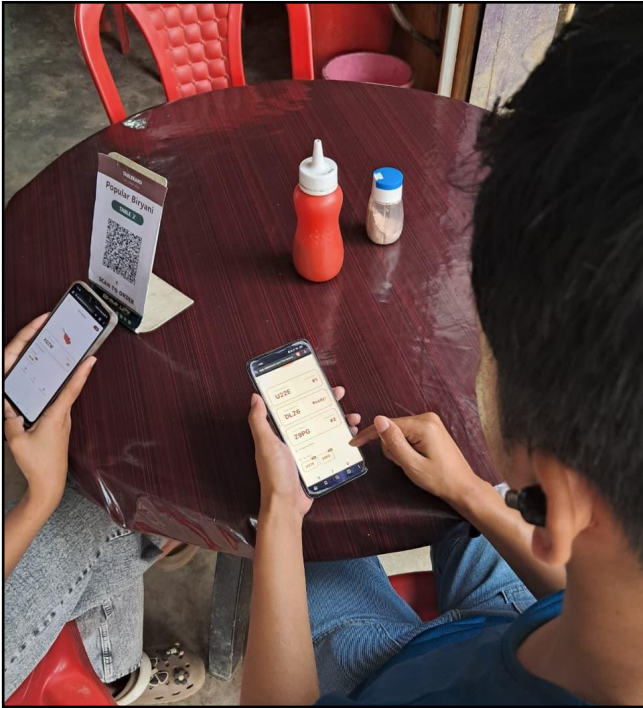


Fig: Customer at table

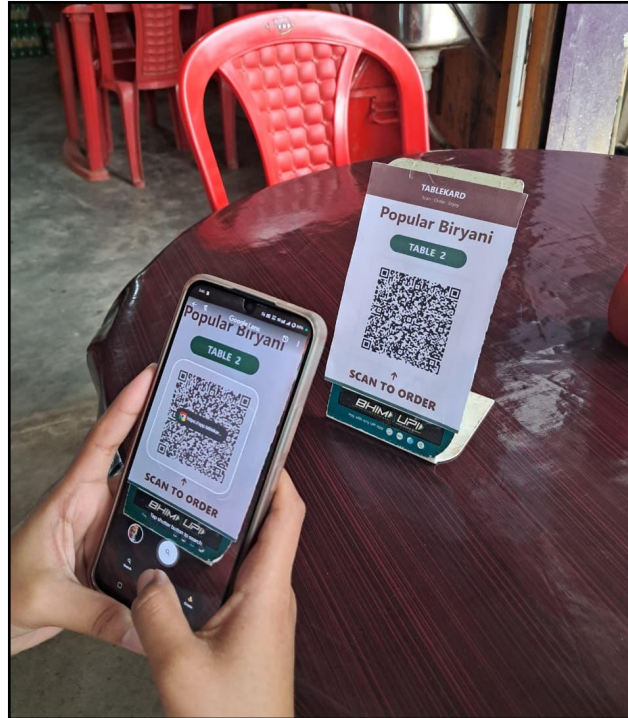


Fig: Customer scans QR Code



Fig: Customer Browse Menu and order food

IMPLEMENTATION AND RESULTS

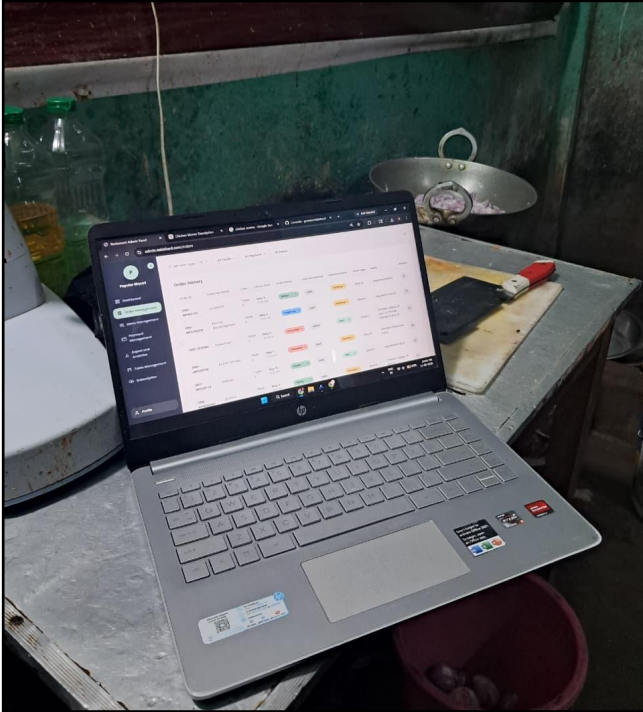


Fig: Admin gets order

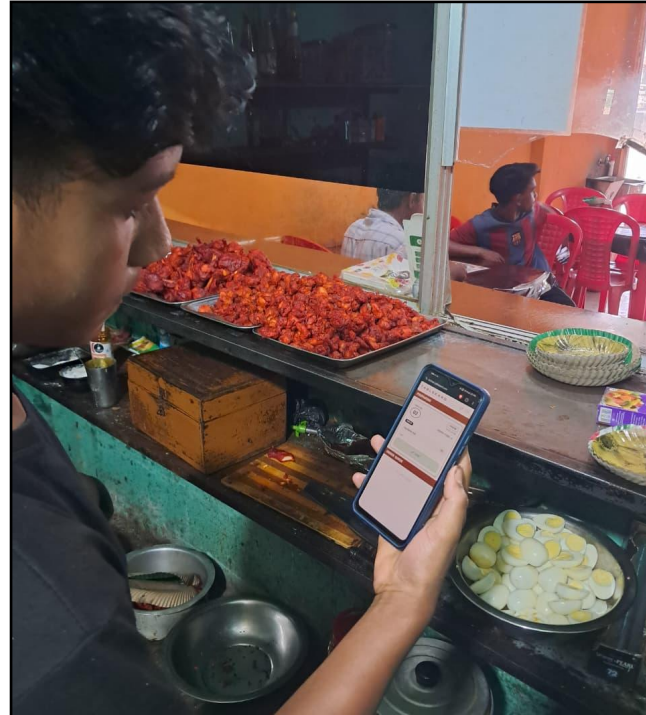


Fig: Cook gets order and prepares it



Fig: customer receives food

CONCLUSION

Tablekard successfully delivers a comprehensive digital transformation solution for the restaurant industry through a modern, scalable SaaS architecture:

1. **Eliminated traditional bottlenecks** – Replaced manual order-taking and standalone POS systems with QR-based digital ordering
2. **Affordable for all restaurant sizes** – Multi-tenant subscription model eliminates heavy upfront costs
3. **Real-time operational efficiency** – WebSocket-based live order sync from customer to kitchen within seconds
4. **Secure and isolated** — PostgreSQL Row-Level Security ensures complete data privacy between restaurant tenants
5. **Integrated payments** — Razorpay gateway with online pay, retry flows, webhook verification, and pay-at-counter fallback
6. **Production-ready tech stack** — Built on React 19, Vite 5, Supabase, and deployed on Vercel for high availability

FUTURE SCOPES

Sl No.	Feature	Description
1	AI & Automation	AI-based menu generation from dish images, predictive sales forecasting, inventory insights
2	Ordering Experience	Voice-enabled ordering for accessibility, smart table reservation with admin sync
3	Restaurant Operations	IoT printer integration for instant kitchen slips, dynamic pricing based on demand and peak hours

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THANK YOU